



P27

Causal Validity of Large-Scale Social Simulation: Phenomenon Reproduction Is Not Mechanism Verification

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Causal Validity of Large-Scale Social Simulation: Phenomenon Reproduction Is Not Mechanism Verification

Working Title (EN): Causal Validity of Large-Scale Social Simulation: Why Phenomenon Reproduction Is Not Mechanism Verification, and Where UDOS RCT Sits on the Validity Ladder

Volume paper P27 • the methodology dialogue paper of P3 "Fault governance RCT" and P20 "Heterogeneity dividend" • UDOS Reasoning Engine v7.6.0 • Fang Wenxin • 2026-09-21

One sentence first: million-scale Agent social simulations such as OASIS, AgentSociety can "reproduce" known social phenomena like group polarization and information diffusion, but "reproducing a phenomenon" and "proving the causal mechanism behind the phenomenon" are two different things—the former only shows your simulator looks like, the latter needs controlled comparison. The reason UDOS's RCT (P3) deliberately keeps fault injection and comparison arms is precisely to push simulation from "phenomenon reproduction" to the validity level of "hypothesis testing."

Evidence caliber (stated only once in the whole paper): this paper is a methodology/validity analysis paper, containing no new UDOS measured numbers; cited UDOS conclusions all go back to v7.5.0 P3/P9. External literature is verified per EVIDENCE_LEDGER_v76.md: OASIS (arXiv:2411.11581, NeurIPS 2024) million-user simulation, reproducing three social phenomena is a verified fact, but its evidence nature is **phenomenon reproduction**; AgentSociety (arXiv:2502.08691) 10,000+ Agents, about 5 million interactions is verbatim verified; MegaAgent (ACL 2025 Findings, 2025.findings-acl.259) 590 Agents national-level policy simulation, making Gomoku in 800 seconds is verbatim verified. AgentTorch / On the Limits of Agency are cited only by "direction/conference existence," their concrete scale numbers not verbatim verified in this volume, not quantitatively cited.

Abstract

2024–2026, LLM-driven large-scale social simulation (Agent-Based Model, ABM) entered the million scale: OASIS supports up to million-user simulation and reproduces information diffusion, group polarization, herding; AgentSociety ran 10,000+ Agents, about 5 million interactions; MegaAgent under no predefined SOP expanded autonomous multi-Agent to 590 for national-level policy simulation. This paper proposes a methodological distinction often masked by such works' "scale narrative": **phenomenon reproduction $\neq$ mechanism verification**. Reproducing a known phenomenon (polarization, herding) only shows the simulator's output is statistically shaped like real data, not that the causal mechanism producing the phenomenon is Agent interaction itself—because in ABM the Agent behavior rules themselves may have been "calibrated" to reproduce that phenomenon. This paper gives ABM's three-level validity ladder: **exploratory level** (discovering emergent phenomena), **phenomenon reproduction level** (reproducing known statistical laws), **hypothesis testing level** (identifying causal effects under controlled comparison), and argues UDOS P3's RCT (fault injection, comparison arms, circuit breaker reassignment) sits at the highest level. Finally points out social simulation's correct use for UDOS: it is a hypothesis generator, not a hypothesis verifier; using it as a verifier is the most dangerous leap in policy inference.

Keywords: Agent-Based Model; social simulation; phenomenon reproduction; causal identification; validity grading; hypothesis testing; RCT

Structured Abstract (background problem → argument → evidence → contribution)

- **Background problem:** million-scale Agent social simulation is increasingly cited as "evidence" by policy inference. But between "the simulator ran out polarization" and "polarization is caused by X" lies the whole causal inference methodology.
- **Core argument:** ABM validity has three levels—exploration / phenomenon reproduction / hypothesis testing; 2026 mainstream million-scale social simulation (OASIS/AgentSociety) sits at the first two, providing no causal identification; UDOS P3's RCT is the third.
- **Evidence clues:** (1) OASIS reproducing three phenomena (paper report, phenomenon reproduction nature); (2) AgentSociety 10,000 Agents/5 million interactions; (3) MegaAgent 590 Agents policy simulation; (4) the basic causal inference distinction (correlation vs intervention).
- **Contribution:** explicitly grading ABM validity into three levels; pointing out the circular-reasoning risk of "reproduction = verification"; drawing for UDOS the division "simulation = hypothesis generation, RCT = hypothesis verification."

1 Introduction: Reproducing a Phenomenon, What Does It Prove

A social simulator ran out "information diffusing in the Agent network like a real social platform, group polarization appearing." This sounds like a strong result. But what does it actually prove?

Which of two things does it prove? (a) This simulator **can reproduce** the statistical shape of known social phenomena; or (b) this simulator **reveals** the causal mechanisms behind these phenomena?

These two are methodologically far apart. (a) is fitting—you tune Agent behavior rules until they reproduce that phenomenon, even in reverse: because you **know** what polarization looks like, you calibrate rules in that direction, finally of course reproducing polarization. (b) is identification—you must prove in controlled comparison that changing some variable indeed caused the phenomenon to change, while excluding other explanations. 2026's million-scale social simulation, the vast majority of honest papers only claimed (a), but citers often read it as (b). This paper makes this leap explicit.

2 What 2026's Large-Scale Social Simulation Is Doing

2.1 Three Verified Representative Works

Per EVIDENCE_LEDGER_v76.md (2026-09-21):

- **OASIS** (arXiv:2411.11581, NeurIPS 2024): based on the social media paradigm, supporting dynamic social networks of up to million users, diverse action spaces and recommendation systems; reproduced three phenomena—information diffusion, group polarization, herding. Its evaluation nature is **phenomenon reproduction**.
- **AgentSociety** (arXiv:2502.08691): 10,000+ Agents, about 5 million inter-Agent/Agent-environment interactions, studying four social issues such as polarization, diffusion.
- **MegaAgent** (ACL 2025 Findings, 2025.findings-acl.259): no predefined SOP, dynamically generating Agents and communication structures, expanding to 590 Agents in national-level policy simulation, and independently making the Gomoku game in 800 seconds.

Additionally, AgentTorch (AAMAS 2025) proposes the Large Population Models (LPMs) paradigm, On the Limits of Agency (AAMAS 2025 Oral) studies the limits of Agent autonomy—this volume cites only by direction, not verbatim citing their scale numbers.

2.2 Their Common Evidence Nature

The common point of these three works is: they show "LLM-driven Agent clusters at large scale can run out presentable social dynamics." This is important **exploratory** capability—previously ABM could only do a few hundred Agents with hand-written rules, now using LLM behavior can do tens of thousands. But their answer to "why this phenomenon appears" usually stays at "because our Agents interact by these rules," rather than "because variable X was proven by controlled comparison to cause Y."

3 The Three-Level Validity Ladder



Figure 1 Exploratory level (discovering emergence) → phenomenon reproduction level (reproducing known statistical laws) → hypothesis testing level (controlled comparison identifying causality). OASIS/AgentSociety sit at the first two, UDOS P3 RCT sits at the third. Conceptual illustration.

This paper explicitly divides ABM validity into three levels:

Table 1 Three levels of ABM validity

Level	Question asked	Representative work	Can answer "why"
L1 Exploratory	What will this set of Agents emerge?	MegaAgent policy simulation	no, only describes emergence
L2 Phenomenon reproduction	Can it reproduce known phenomena?	OASIS reproducing three phenomena, AgentSociety	partially, proves fitting looks like
L3 Hypothesis testing	Does changing X causally cause Y?	UDOS P3 fault-injection RCT	yes, controlled comparison identifies the effect

3.1 L1→L2: From "Moves" to "Looks Like"

L1 only asks what emergent behavior the system ran out. MegaAgent under no SOP generating 590 Agents and doing policy inference, writing Gomoku in 800 seconds, this itself is proof of emergent capability, but it is not responsible for "whether this policy inference is right."

L2 requires output matching the statistical shape of real data. OASIS compares simulated output with real social media data, reproducing the distribution features of three phenomena—this is stronger than L1, but it answers "looks like or not," not "why."

3.2 L2→L3: From "Looks Like" to "Causal"—This Is the Hardest Leap

L3 requires controlled intervention: holding other conditions fixed, changing only one variable, observing whether the outcome systematically changes, and giving confidence intervals. This is the basic requirement of causal inference, also the weakest link of LLM-ABM—because ABM's "experiment" itself is done in a world where you can tune parameters arbitrarily, you can fully make results go your desired direction by tuning.

The key risk is circular reasoning: if when calibrating Agent rules you already referenced the real-world polarization distribution, then "the simulator reproducing polarization" is not an independent discovery, but you writing the answer into parameters. This is not saying OASIS-like works are cheating, but its evidence strength cannot exceed "a well-fitted phenomenon reproducer."

3.2.1 Three "Looks Like an Experiment" Pseudo-L3

In practice, three practices are often mistaken for causal identification, must be exposed:

1. **Post-tuning fitting = causality.** Tuning Agent behavior rules until they match the real data distribution, then claiming "the model explains polarization." This is only fitting, the rules themselves may not be polarization's real mechanism at all.
2. **Single-arm demonstration = effect.** Running one large-scale simulation, showing a dramatic emergence (such as group split), then claiming "some variable causes split." No comparison group, cannot exclude randomness or initial-condition sensitivity.
3. **Sensitivity analysis = robustness.** Sweeping a few parameters to see whether emergence still appears, then claiming the conclusion robust. Sensitivity analysis only shows "changing parameters the phenomenon remains," not "changing some variable indeed caused the effect."

Real L3 requires **comparison design + repeatability + confidence intervals**, UDOS P3's fault-injection comparison arms precisely align with these three.

3.2.2 External Validation: Anchoring Simulation to the Real World

One way out of circular reasoning is externally validating simulation results against real-world **quasi-experiments** or historical policy comparisons. If after changing some variable the LLM-ABM's predicted effect direction and magnitude agree with a real-world natural experiment's estimate, it moves up from L2 to near L3. But 2026's mainstream million-scale social simulation mostly has not done this external anchoring—their "validation" still stays at "consistent with historical statistical shape," rather than "consistent with the real causal effect of an intervention." This is precisely why this paper overall places them at L2.

4 Why UDOS P3's RCT Sits at L3

4.1 The RCT's Comparison Group and Intervention

UDOS P3 (fault governance RCT)'s design deliberately aligns with L3 requirements: it is not "running a multi-Agent system once to see results," but **comparison arms under fault injection**—in an end-to-end task of 24 work orders, 9 specialists, 7 QA (including 2 Byzantine), 3 fault specialists, the layered hybrid topology with circuit breaker reassignment achieves 24/24 closing, all 3 fault nodes isolated, 5 reworks (cpu-pto). Here the "intervention" is injecting faults, the "comparison" is different topology/circuit

breaker strategies, the "outcome" is completion rate. It wants to answer "whether circuit breaker reassignment **causes** completion rate to rise," not "whether the system running looks like."

4.2 Simulation vs RCT Division

Thereby obtaining UDOS's correct positioning of social simulation:

- **Large-scale social simulation (OASIS/AgentSociety) = hypothesis generator.** It cheaply tells you "in this region of parameter space, what interesting phenomena emerge," helping you propose hypotheses worth testing.
- **Controlled RCT (P3) = hypothesis verifier.** It expensively tells you "under strict comparison, whether this intervention really caused that outcome."

Taking simulation as a verifier is the most dangerous leap in policy inference—you will mistake "I saw X in the simulator" for "X will happen in the real world."

5 Echo with P20: Phenomenon Reproduction Does Not Identify Causality Between Individuals

P20 already pointed out, OASIS observes "larger Agent groups bring stronger group dynamics and more diverse opinions," but this is phenomenon reproduction, not answering "how much increasing N causes opinion diversity to change." This paper generalizes this: OASIS-like works can show "scale and diversity correlated," but to jump from correlation to "scale **causes** diversity" needs L3 controlled comparison—e.g. fixing other conditions, changing only the Agent count, measuring systematic changes in opinion distribution. OASIS did not do this, so its "larger more diverse" is observation, not causal effect estimation.

Worth adding the methodological why: ABM is inherently "forward simulation"—you input rules, it outputs behavior. Reverse-inferring from output "which rule caused which behavior" under multi-Agent nonlinear interaction is ill-posed, because rules entangle with each other, initial conditions are sensitive, emergence is nonlinear. To solve this ill-posed problem, the only clean way is forwardly doing controlled intervention: moving one rule at a time, seeing whether the outcome moves. This is precisely the fundamental division of RCT and ABM—ABM is responsible for forward exploration of "what happens," RCT is responsible for counterfactual verification of "was it this thing that caused it."

Pushing this division into policy inference practice: using OASIS-like simulation to scan out "under some policy parameter group split may appear," this is a valuable risk warning; but based on it pushing that policy parameter all the way in some direction and claiming "this avoids split," crosses the validity boundary—because the split in simulation may come from the Agent rules you wrote, not the policy itself. The safe practice is taking the direction simulation hints, going back to the real world to find a quasi-experiment or small-scale pilot for L3 validation.

6 Drawbacks, Failure Boundaries, and Falsifiable Conditions

6.1 Drawbacks and circulating exaggerations

1. **"Reproduced polarization" read as "proved the polarization mechanism."** This is ABM narrative's most common circular reasoning: the rules are written by you, the phenomenon is what you want to reproduce, reproduction success does not constitute independent evidence.

2. **"Million scale" read as "conclusion more credible."** Large scale only raises the richness of emergent phenomena, not the strength of causal identification; a million-scale well-fitted phenomenon reproducer's evidence strength remains lower than a few-dozen-node controlled RCT.
3. **Taking simulation output directly as policy advice.** Before doing L3 controlled comparison, social simulation can only "hint risk direction," cannot "give policy dosage."

6.2 Failure boundaries

- This paper does not deny large-scale social simulation's exploratory value; it is a sharp tool for hypothesis generation. This paper only draws its evidence applicability boundary for **causal policy conclusions**.
- If some future ABM work under controlled intervention gives causal effect estimates consistent with real-world quasi-experiments, then that work crosses into L3, this paper's "ABM mostly at L2" judgment needs upgrading.

6.3 Falsifiable conditions

- **C1:** if OASIS/AgentSociety-like works are proven their Agent behavior rules calibrated independently of the target phenomenon (i.e. not referencing polarization distribution for tuning), and output consistent with real data in held-out periods, then their evidence strength can be upgraded from L2; conversely if rules are found repeatedly tuned to reproduce phenomena, then circular reasoning risk is confirmed.
- **C2:** if a purely L1/L2 social simulation is used to give policy-binding "dose-effect" conclusions (rather than directional hints), and afterward falsified by real policy implementation, then this paper's "simulation $\neq$ verification" demarcation is strengthened.

7 Hard Predictions (2026–2031)

- **F1:** in the next five years, the LLM-ABM field will appear review standards specifically distinguishing "phenomenon reproduction" and "causal identification"; papers still taking reproduced phenomena directly as policy evidence will be systematically downgraded.
- **F2:** the first LLM-ABM conclusion accepted by the policy community must simultaneously attach an L3-level controlled comparison or external validation with a real-world quasi-experiment; purely phenomenon-reproducing "million-Agent policy inference" will no longer alone constitute a decision basis.

8 Conclusion

OASIS, AgentSociety, MegaAgent pushed LLM social simulation from a few hundred rule Agents to tens of thousands of LLM Agents, this is real progress. But progress happens at "how large a scale can be simulated," not at "how hard a causality can be identified." Reproducing a phenomenon only shows the simulator tuned to look like, not that you found the mechanism behind the phenomenon—the latter needs controlled intervention, comparison arms, confidence intervals, i.e. UDOS P3-style RCT.

For engineering, this means large-scale social simulation's correct position in the UDOS system is **hypothesis generation**: it cheaply scans parameter space, hints directions worth testing; upgrading directional hints into decidable causal conclusions must return to RCT. For research, it gives a killable standard (C1–C2): any ABM claiming mechanism discovery must prove its rules were not reverse-calibrated by that mechanism's target phenomenon.

The simulator can run out the world's appearance, which does not mean the simulator found the world's cause. Taking the former as the latter is replacing experiment with fitting.

References

2026 frontier material (existence verified per EVIDENCE_LEDGER_v76.md 2026-09-21; numbers all paper reports)

1. Yang, et al. (2024). OASIS: Open Agent Social Interaction Simulations with One Million Agents. arXiv: 2411.11581, NeurIPS 2024.
2. Piao, et al. (2025). AgentSociety: Large-Scale Simulation of LLM-Driven Generative Agents. arXiv: 2502.08691.
3. (MegaAgent Team) (2025). MegaAgent: A Large-Scale Autonomous LLM-based Multi-Agent System Without Predefined SOPs. ACL 2025 Findings, 2025.findings-acl.259.
4. AgentTorch / Large Population Models. AAMAS 2025. (direction citation, scale numbers not verbatim verified)
5. On the Limits of Agency in Agent-based Models. AAMAS 2025 Oral. (direction citation)

UDOS paper volumes (this series, evidence grades marked in text)

1. UDOS v7.5.0: P3 "Circuit breaker reassignment RCT for multi-agent fault governance," P9 "Quality saturation law of the Agent legion."
2. UDOS v7.6.0: P20 "Heterogeneity dividend and homogeneity saturation," P21 "Million-level coordination end-to-end evidence and extrapolation boundaries."

Evidence Discipline and Reproduction Notes

- This paper is a methodology/validity analysis paper, no new experiments; UDOS numbers go back to P3/P9 and `reports7/`, evidence grades subject to each paper's marking.
- OASIS/AgentSociety/MegaAgent existence and scale numbers are verified per EVIDENCE_LEDGER_v76.md; AgentTorch and On the Limits of Agency are cited only by direction, not verbatim citing scale numbers. All external numbers keep the "paper report" caliber.
- This paper's predictions (F1–F2) and falsification conditions (C1–C2) can be post-hoc scored with public evidence during 2026–2031.